

MSA Training Guide - From Awareness to Expert

Structured 3-level MSA training curriculum | AIAG MSA 4th Edition

QUALITY HEAD AGENTS | AIAG MSA 4th Edition | IATF 16949:2016

Level 1 - MSA Awareness (Operators, Technicians)

- Duration: 2 hours
- Why measurement matters: 'Garbage in = garbage out' - bad measurement = wrong decisions
- What is repeatability: 'Will this gauge give the same answer if I measure twice?'
- What is reproducibility: 'Will Operator A and Operator B get the same answer?'
- What is calibration and why expired calibration = stop using the gauge immediately
- How to conduct your part of a GRR study: measure parts consistently, one at a time, record carefully
- What to do if you see gauge drift or unusual readings: report immediately, do not continue

Level 2 - MSA Practitioner (Engineers, QA Staff)

- Duration: 1 day
- AIAG MSA 4th Edition framework: 5 study types, when each is required
- GRR study design: selecting operators, parts, trials. Randomization. Blinding.
- GRR calculation: EV, AV, GRR, PV, TV, NDC - manual and software methods
- Interpreting results: % Study Var vs % Tolerance - which to report when
- Bias study: conducting t-test, interpreting statistical significance
- Linearity study: regression method, R^2 interpretation
- Stability study: control chart setup, interpreting X-bar and R chart for gauge drift
- Attribute GRR: Kappa coefficient, 50-part study design
- PPAP Element 8: what to submit, how to present MSA results to customer

Level 3 - MSA Expert / Facilitator (Quality Head / Manager)

- Duration: 2 days + practical assessment
- Advanced ANOVA method vs Average and Range method - when each applies
- Expanded uncertainty and 95% confidence intervals for MSA results
- Customer-specific MSA requirements: Ford Q1, GM IATF, Stellantis MFGP
- Conducting MSA cross-functional review and sign-off process
- Developing plant-wide MSA Plan and Gauge Register management system
- Training and assessing Level 1 and Level 2 staff
- Audit simulation: respond to MSA audit questions as Quality Manager
- IATF 16949 clauses 7.1.5.1, 7.1.5.2, 8.3.4.4, 8.5.1.1 deep dive